

Unit 101 Field Data Review

Mohawk Valley West Slope demonstration. **Tractor** unit, 97.4 ac. 45 plots, BAF 20, variable-radius cruise on a 300 ft grid, 478 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	212 sq ft/ac (1 SE of mean, BA 1.9 %)	Type	Sierran mixed conifer
Trees per acre	123	CWHR size / density	4 / D (cover 83 %)
QMD	17.8 in	SDI	288 (51 % of max 567)
SAWTIMBER (≥ 10 in)		BIOMASS (< 10 in)	
BA QMD	203 sq ft/ac 19.8 in	BA QMD	9 sq ft/ac 7.6 in
TPA	95	TPA	28
Volume	29.8 MBF/ac (5943 cu ft)	Volume	2.9 green tons/ac (130 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 146 sq ft/ac BA	Vol. sampling error	10.4 % at 95 % (CV 35 %, t = 2.02, n = 45)
Species (BA)	PP 70, WF 60, IC 32, DF 29, SP 21	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 6
		Sale volume share	4.4 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	3.9 % at 95 % (CV 13 %); reference, not the standard
		Measured trees	PP 158, WF 134, DF 66, SP 46, IC 71; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	45; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	8	15.6	3.6	51	0.3
8-11.9	34	27.7	15.1	241	1.2
12-15.9	74	30.7	32.9	670	3.4
16-19.9	82	21.2	36.4	870	4.4
20-23.9	68	11.7	30.2	831	4.2
24-27.9	50	6.0	22.2	686	3.4
28-31.9	52	4.8	23.1	782	3.9
32-35.9	33	2.3	14.7	534	2.7
36-39.9	25	1.4	11.1	439	2.2
40-43.9	18	0.9	8.0	310	1.6
44-47.9	10	0.4	4.4	173	0.9
48-51.9	3	0.1	1.3	59	0.3
52-55.9	9	0.3	4.0	189	0.9
56-59.9	4	0.1	1.8	88	0.4
60-63.9	7	0.2	3.1	151	0.8
All	477	123.4	212.4	6073	30.5

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	158	45.2	70.2	16.9	2158	10.8
WF	134	31.5	59.6	18.6	1701	8.5
IC	71	19.1	31.6	17.4	768	3.9
DF	66	14.3	29.3	19.4	849	4.3
SP	47	13.3	20.9	17.0	574	2.9
PPP	1	0.1	0.4	35.1	23	0.1
All	477	123.4	212.4	17.8	6073	30.5

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 101 plots, 300 ft grid (orange: QA flag)

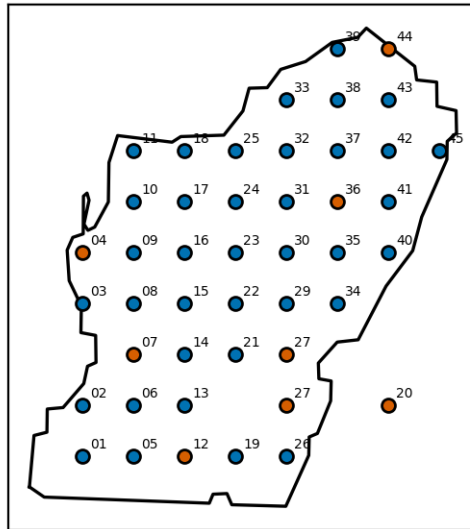


Figure 1. Plot locations with QA status.

QA findings: 7 item(s) for crew follow-up

Plot	Finding
101-04	Tree 1: species code 'PPP' not valid. Correct in edit.
101-07	Tree 9: height 32 ft outside tolerance for DBH 27.8. Remeasure.
101-12	Tree 1: DBH 999.0 out of range. Remeasure.
101-20	Plot falls outside the unit boundary. Check GPS position and unit number.
101-27	Plot id recorded 2 times. Check plot numbers.
101-36	Tree 2: height missing. Measure.
101-44	Tree 1: height 15 ft outside tolerance for DBH 34.0. Remeasure.

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
101-01	12	240	11	ok
101-02	10	200	14	ok
101-03	6	120	28	ok
101-04	9	180	25	FLAG
101-05	9	180	16	ok
101-06	12	240	14	ok
101-07	13	260	9	FLAG
101-08	11	220	10	ok
101-09	11	220	19	ok
101-10	12	240	25	ok
101-11	9	180	24	ok
101-12	10	200	8	FLAG
101-13	11	220	5	ok
101-14	10	200	7	ok
101-15	10	200	3	ok
101-16	11	220	6	ok
101-17	11	220	4	ok
101-18	10	200	10	ok
101-19	10	200	8	ok
101-20	11	220	9	FLAG
101-21	11	220	10	ok
101-22	10	200	5	ok
101-23	11	220	3	ok
101-24	11	220	5	ok
101-25	11	220	6	ok
101-26	15	300	15	ok
101-27	11	220	9	FLAG
101-27	9	180	17	FLAG
101-29	12	240	8	ok
101-30	11	220	3	ok
101-31	11	220	3	ok
101-32	11	220	4	ok
101-33	9	180	7	ok
101-34	9	180	9	ok
101-35	10	200	2	ok
101-36	12	240	3	FLAG
101-37	11	220	6	ok
101-38	11	220	14	ok
101-39	10	200	4	ok
101-40	12	240	1	ok
101-41	12	240	2	ok
101-42	10	200	4	ok
101-43	9	180	8	ok
101-44	11	220	6	FLAG
101-45	10	200	10	ok